RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Electronics and Communication Engineering

Academic Year: 2023 - 2024 (Odd Semester)

Name of the Faculty member: Mrs.V.SrirengaNachiyar
Course Code & Title : CBM370 & Wearable Devices
Semester, Degree & Branch : V B.E ECE A & B
Topic : Applications of Wearable Systems
Innovative Method : One Minute Paper

Wearable Devices 953621106015

1) Twave, ~~P~~ wave, QRS wave
2) Role of wearables
3) Piezo Electric sensor, Transducer
4) Alpha, Beta, Teta
5) Garment sensor, Local processing, Conditioning, filtering of the symbols
6) Public key, secure socket layer, infrastructure

Reg no: 953621106012

1) T wave, P-wave, QRS wave
2) Role of wearables
3) Piezo electric sensors, transducers
4) Alpha, Beta, Teta
5) · Garment sensor
   · Local processing
   · Conditioning. filtering of the symbols
6) secure socket layer
   · public key infrastructure

A Ruhaina Thasneem
953621106084

1. PR, QRS, RT
2. Role of Wearables.
3. Transducer.
4. Beta, Alpha, Theta, Delta.
5. Garment and Sensor, Conditioning & filtering of the Signals, local processing.
6. PKI, SSL.

1) ECG
   1. P wave
   2. QRS complex
   3. T wave
2) Role of wearable
3) Piezoelectric transducer
4) EEG → Beta, Alpha, Theta, Delta
5) Sensor layer → lowest layer: skin, filtering of signals, local processing
6) Public key, Secure socket layer

Signature of the faculty member 16/8/23

HOD 19/8/23

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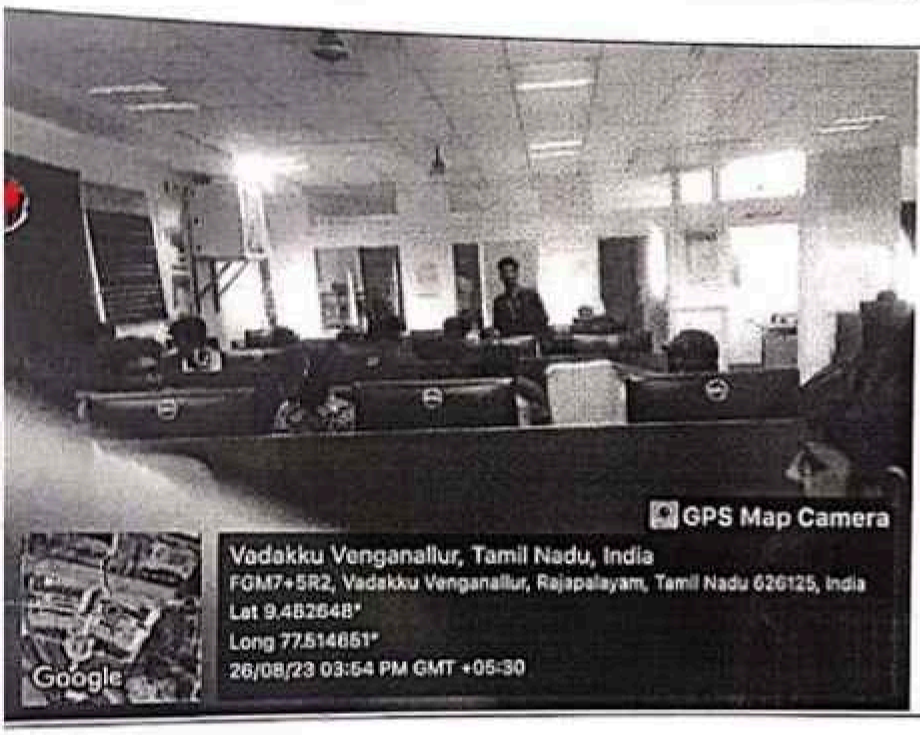
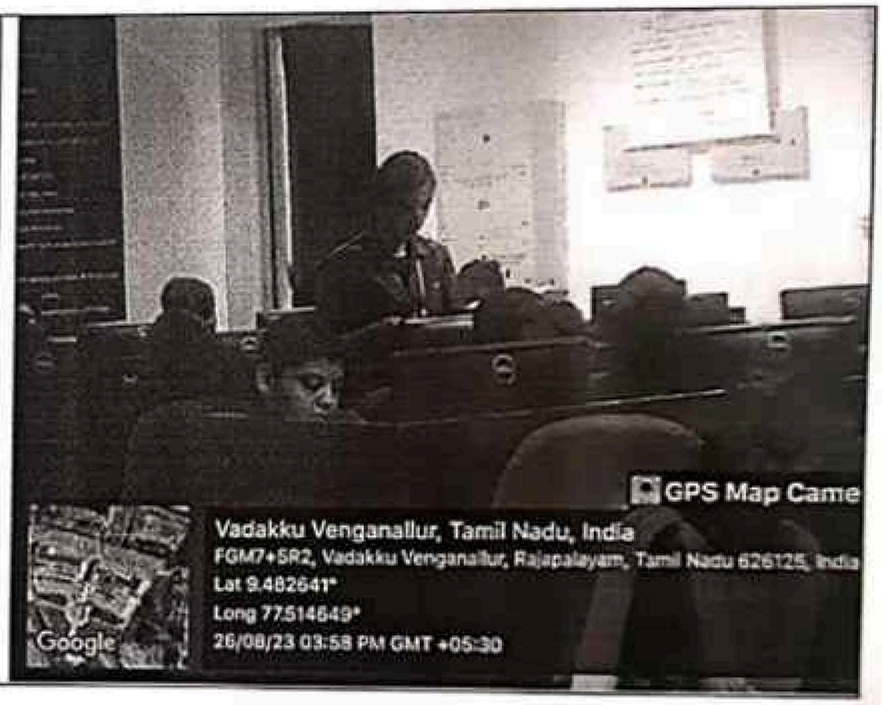
Department of Electronics and Communication Engineering
Academic Year: 2023 - 2024 (Odd Semester)

Name of the Faculty member: Mrs.V.SrirengaNachiyar
Course Code & Title : CBM370 & Wearable Devices
Semester, Degree & Branch : V B.E ECE A & B
Topic : Human body as a heat source for power generation
Innovative Method : Zero Minute Speech





28/8/23
Signature of the faculty member

HOD

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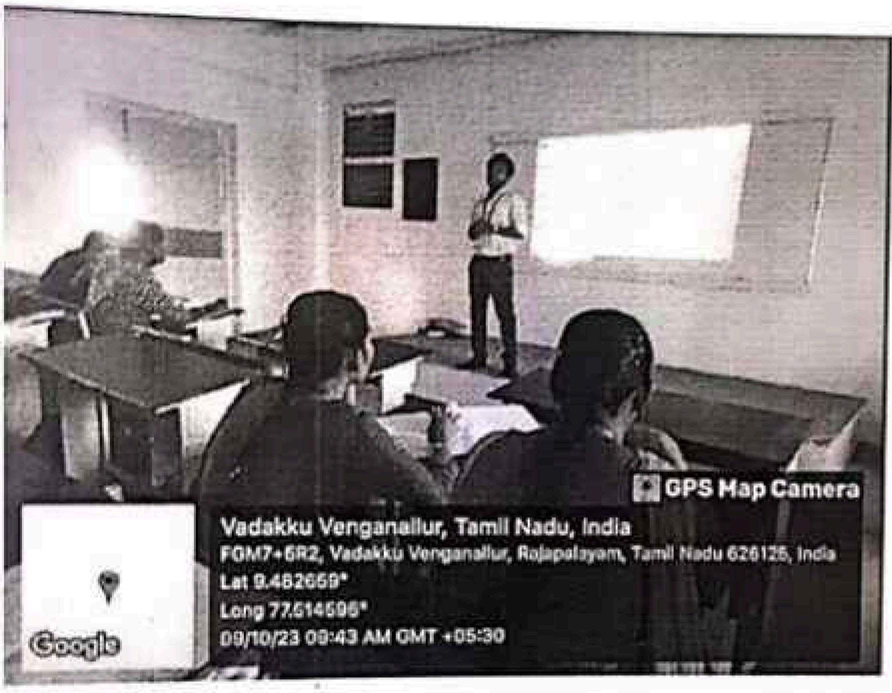
Department of Electronics and Communication Engineering
Academic Year: 2023 - 2024 (Odd Semester)

Name of the Faculty member: Mrs.V.SrirengaNachiyar
Course Code & Title : CBM370 & Wearable Devices
Semester, Degree & Branch : V B.E ECE A & B
Topic : Wireless communication Techniques
Innovative Method : Brain Storming



10/10/23
Signature of the faculty member

10/10/23
HOD

Department of Electronics and Communication Engineering
Academic Year 2023 – 2024 (Odd Semester)
Degree, Semester & Branch: V Semester B.E. ECE B
Course Code & Title: CBM370 Wearable Devices
Name of the Faculty member (s): Mrs.V.SrirengaNachiyar

Innovative Practice Description

- **Unit / Topic: Unit IV** / Case study - smart fabric for monitoring biological parameters
- **Course Outcome:** CO 4
- **Unit Outcome:** 10
- **Activity Chosen: Jigsaw**
- **Justification:**

  Jigsaw is a cooperative group activity in which students are interdependent to achieve a common goal. In part one, each group is provided a different problems in Input /Output Quantization Errors. The group members become experts on that particular topic and create a group response.
- **Time Allotted for the Activity:** 50 Minutes
- **Details of the Implementation:**

  Students said that they felt this activity had improved their understanding of the smart fabric for monitoring biological parameters. Because in this unit, topic is case study. Some changes will be implemented for the particular topic. During assessment tests, it is simple to recall the case studies related to smart fabric for monitoring biological parameters.
- **CO – PO / PSO mapping:**

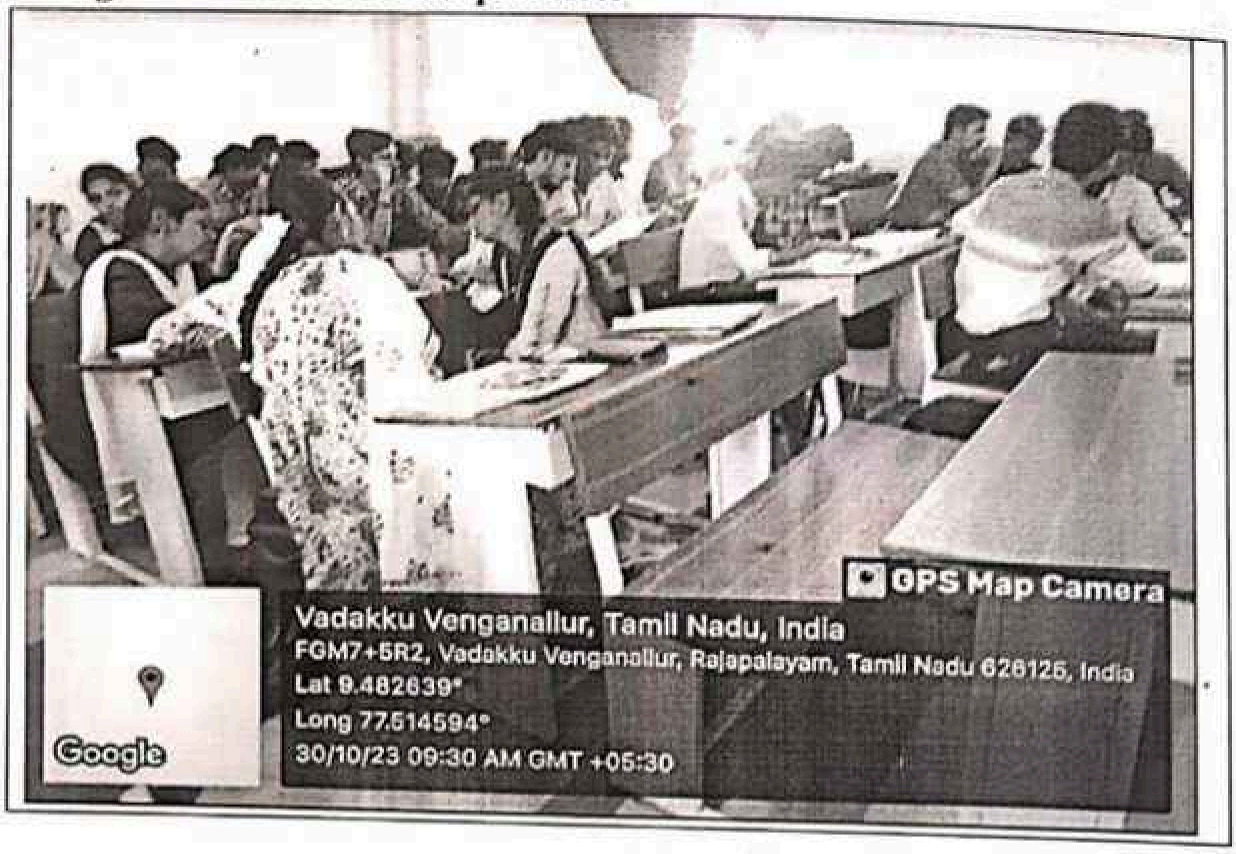
CO	PO1	PO2	PO3	PO4	PO10	PO12	PSO2
CO4	3	2	2	1	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO10
	2
Justification for correlation	Students discussed with their peers and presented the case studies related to smart fabric to measure the biological parameters for the person under observation in hospital. So, it is mapped at level 2

- **Images / Screenshot of the practice:**



- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***

    Students stated that they felt that this activity helped them to understand the concept of smart fabric and its usefulness to measure the biological parameters like Heart rate, pulse rate, etc. It is very useful to recollect the important points while writing examination

  - ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

    The Students eagerly participated in that activity and then they acquired the knowledge about various parameters to be measured by wearing smart fabrci. Due to this activity, the students were able to recall the topic and remember the important terms easily.

  - ***Challenges faced in implementation:***

    Students took some time to forming the group. After formed the group, they discussed the topic and present the important terms. I encouraged the students for completing the activity with useful points.

References:

- Annalisa Bonfiglo and Danilo De Rossi, Wearable Monitoring Systems, Springer, 2011.

21/11/23

Signature of the faculty member

21/11/23

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Department of Electronics and Communication Engineering
Academic Year 2023 – 2024 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. ECE B
Course Code & Title: CBM370 Wearable Devices
Name of the Faculty member (s): Mrs.V.SrirengaNachiyar

Innovative Practice Description

- **Unit / Topic: Unit V /** Sports Medicine
- **Course Outcome:** CO 5
- **Unit Outcome:** 13
- **Activity Chosen: Flipped class**
- **Justification:**

The flipped classroom model is based on the idea that traditional teaching is inverted in the sense that what is normally done in class is flipped or switched with that which is normally done by the students out of class. This Activity gives the clear understanding of the particular topic.

- **Time Allotted for the Activity:** 30 Minutes
- **Details of the Implementation:**

Students were given the task of preparing a presentation on the topic of " Sports Medicine ". They described about the wearable systems used in sports medicine". Students were divided into groups and each group will prepared the detailed presentation of sports medicine and deliver their presentation among all the students.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO4	PO9	PO10	PO12	PSO3
CO5	2	2	3	1	2	2	1	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO10	PO12
	2	1
Justification for correlation	Students will understand the wearable devices used in sports and will give clear information during discussion session. Hence it is mapped at level 2	Students were identify the previous instances in wearable system and how it can be used where improvements in technology required lifelong learning for practitioners in accordance with changing. Thus level 1 is mapped

Reflective Critique:

- ***Feedback of practice from students and other stakeholders:***

  The practise was well-received by students, who said it helped them remember subjects and substance while writing exams. They also noted that because they are presenting in front of all of the students, this activity boosts their confidence.

- ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

  The Students eagerly participated in that activity and then they acquired the knowledge of various wearable devices used in sports. Due to this activity, the students were able to recall the topic and remember the important terms easily.

- ***Challenges faced in implementation:***

  I have planned this activity for 30 minutes. Students take some time to present the topic, they present the topic with presentation. Students asked doubts regarding the topic before presentation. I've given a brief overview of the various wearable systems used for sports and encouraged the students to present which it takes another 5 minutes for completing the activity.

References:

- Annalisa Bonfiglo and Danilo De Rossi, Wearable Monitoring Systems, Springer, 2011.

21/11/23

Signature of the faculty member

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